// Dimitrios Vasilopoulos

Postdoctoral Researcher – SnT | University of Luxembourg

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SUMMARY

I am a Postdoctoral Researcher at the Interdisciplinary Centre for Security, Reliability, and Trust (SnT), University of Luxembourg, and a member of the CryptoLUX research group led by Alex Biryukov. My research focuses on blockchain security and privacy, with an emphasis on designing cryptographic protocols with formal security guarantees for modern distributed systems.

Previously, I was a Senior Postdoctoral Researcher at the Insight SFI Research Centre for Data Analytics, University College Cork, where I worked with Paolo Palmieri on privacy-enhancing technologies. Prior to that, I held a postdoctoral position at IMDEA Software Institute in Madrid, collaborating with Pedro Moreno-Sanchez on provably secure, privacy-preserving protocols for blockchain-based payment systems.

I received my PhD in 2019 from EURECOM and Sorbonne University for research conducted under the direction of Refik Molva and Melek Önen. My doctoral research focused on verifiable cloud storage, designing protocols that allow users to verify the integrity and reliability of outsourced data.

I have authored *seven* peer-reviewed publications in leading venues in security and applied cryptography, and have delivered *five* invited and conference talks. I have served on the program committees of *thirteen* international venues, reviewed for *five* journals, and contributed to graduate-level teaching. I have also co-supervised *one* PhD student and *five* Master's students.

EDUCATION

PhD in Information Technology, Telecommunications, and Electronics, July 2019
Sorbonne Université | EURECOM, Sophia Antipolis, France
Advisors: Refik Molva and Melek Önen
Thesis: *Reconciling Cloud Storage Functionalities with Security: Proofs of Storage with Data Reliability and Secure Deduplication*

MSc in Communications and Computer Security, January 2014
Télécom ParisTech | EURECOM, Sophia Antipolis, France

Engineering Diploma in Electronic and Computing Engineering (5–year), March 2010
Technical University of Crete, Chania, Greece

EMPLOYMENT

Postdoctoral Researcher — Sep 2025 – Present
SnT | University of Luxembourg, Esch-sur-Alzette, Luxembourg
Host: Alex Biryukov (CryptoLUX Research Group)
Research area: Cryptography for blockchain security and privacy

- *CryptoFin: Advanced Cryptography for Finance and Privacy* – Luxembourg National Research Fund (FNR). Scientific contributor.
- *FutureFinTech Lighthouse 2: Compliant, transparent and efficient crypto-asset markets* – Luxembourg National Research Fund (FNR). Scientific contributor.

Senior Postdoctoral Researcher — Dec 2024 – Aug 2025
Insight SFI Research Centre for Data Analytics | University College Cork, Cork, Ireland
Host: Paolo Palmieri
Research area: Privacy-enhancing technologies

- *Benchmarking Differential Privacy Through a Generalised Privacy Budget ε* – Meta Platforms, Inc. via Empower Science Foundation Ireland Spoke. Scientific contributor.

Postdoctoral Researcher — Jun 2021 – May 2024
IMDEA Software Institute, Madrid, Spain
Host: Pedro Moreno-Sanchez
Research area: Cryptographic protocols for blockchain-based payment systems

- *PRODIGY: Ensuring the Security, Scalability and Correctness of Digital Provenance Systems* – Spanish Research Agency (AEI). Scientific contributor.
- *BLOQUES: Scalable and Secure Smart Contracts and Blockchains through Verification and Analysis* – Community of Madrid. Scientific contributor.

Doctoral Researcher Mar 2015 – Jul 2019
EURECOM, Sophia Antipolis, France
Advisors: Refik Molva and Melek Önen
Research area: Cryptographic protocols for verifiable and reliable cloud storage

- *TREDISEC: Trust-Aware, Reliable and Distributed Information Security in the Cloud* – H2020-ICT, EU. Scientific contributor and deliverables author.

Research Intern (during MSc, 6 months) Jul 2013 – Jan 2014
SecludIT, Valbonne, France
Mentors: Refik Molva and Sergio Loureiro
Topic: Security auditing of cloud infrastructures

IT Support & Computer Skills Instructor Jan 2011 – Jul 2012
Freelancer, Athens, Greece
Delivered technical support to small businesses and individuals; trained adults in foundational computer skills.

PUBLICATIONS

Refereed Conference and Journal Publications

Conference rankings (A*/A/B) follow the CORE classification at the time of publication.

[1] *Algebraic Zero Knowledge Contingent Payment*.
Javier Gomez-Martinez, Dimitrios Vasilopoulos, Pedro Moreno-Sanchez, and Dario Fiore.
ACNS 2025 (Acceptance rate: ~23%; CORE B).

[2] *MixBuy: Contingent Payment in the Presence of Coin Mixers*.
Diego Castejon-Molina, Dimitrios Vasilopoulos, and Pedro Moreno-Sanchez.
PoPETs 2025 (Acceptance rate: ~26%; CORE A).

[3] *Cryptographic Oracle-based Conditional Payments*.
Varun Madathil, Sri AravindaKrishan Thyagarajan, Dimitrios Vasilopoulos, Lloyd Fournier, Giulio Malavolta, and Pedro Moreno-Sanchez.
NDSS 2023 (Acceptance rate: ~16.4%; CORE A*).

[4] *Proofs of Data Reliability: Verification of Reliable Data Storage with Automatic Maintenance*.
Dimitrios Vasilopoulos, Melek Önen, Refik Molva, and Kaoutar ElKhiyaoui.
Security and Privacy (vol. 6), Wiley, September 2023 (Invited submission, special issue).

[5] *PORTOS: Proof of Data Reliability for Real-World Distributed Outsourced Storage*.
Dimitrios Vasilopoulos, Melek Önen, and Refik Molva.
SECRYPT 2019 (Acceptance rate: ~16%, full papers; CORE B).
➤ Recipient of the Best Student Paper Award.

[6] *POROS: Proof of Data Reliability for Outsourced Storage*.
Dimitrios Vasilopoulos, Kaoutar ElKhiyaoui, Refik Molva, and Melek Önen.
ACM SCC@ASIACCS 2018 (Acceptance rate: ~35%).

[7] *Message-Locked Proofs of Retrievability with Secure Deduplication*.
Dimitrios Vasilopoulos, Melek Önen, Kaoutar ElKhiyaoui, and Refik Molva.
ACM CCSW@CCS 2016 (Acceptance rate: ~29%, full papers).

Preprints and Submitted Work

- *Geo-Indistinguishability in Practice: Mitigating Practical Location De-Anonymisation Using Differential Privacy*.
Akasha Shafiq, Hamza Aguelal, Dimitrios Vasilopoulos, and Paolo Palmieri.
2026. (Under submission)
- *Cryptographic Collateralized Loan without Smart Contract*.
Diego Castejon-Molina, Varun Madathil, Dimitrios Vasilopoulos, Sri AravindaKrishan Thyagarajan, and Pedro Moreno-Sanchez.
Cryptology ePrint Archive, 2026/1123. (Under submission)

- *Foundations of Verifiably Encrypted (Blind) Signatures*.
  Diego Castejon-Molina, Erkan Tairi, Dimitrios Vasilopoulos, and Pedro Moreno-Sanchez.
  Cryptology ePrint Archive, 2026/789. (Under submission)
- *The Effectiveness of Differential Privacy in Real-world Settings: A Metrics-based Framework to help Practitioners Visualise and Evaluate ε*.
  Akasha Shafiq, Abhishek Kesarwani, Dimitrios Vasilopoulos, and Paolo Palmieri.
  Cryptology ePrint Archive, 2025/1173. (Under submission)
- *A Cryptographic Layer for the Interoperability of CBDC and Cryptocurrency Ledgers*.
  Diego Castejon-Molina, Alberto del Amo Pastelero, Dimitrios Vasilopoulos, and Pedro Moreno-Sanchez.
  Cryptology ePrint Archive, 2023/116.

RESEARCH FUNDING

Grant Proposals (Under Review)

- *FCRYPT: Next-Generation Cryptography for Future Security and Privacy-Enhancing Technologies* – FNR CORE, 2026 (PI: Alex Biryukov). Major contributor to proposal writing and technical design (non-PI).

TEACHING EXPERIENCE

Teaching Assistant
Department of Computer Science, University of Luxembourg, Esch-sur-Alzette, Luxembourg

- MSc in Information and Computer Sciences – Course: *Advanced cryptographic algorithms and blockchain cryptography*, Fall 2025 (Instructors: Alex Biryukov and Jean-Sébastien Coron). Supervised student projects.

Teaching Assistant & Guest Lecturer
School of Computer Science & IT, University College Cork, Cork, Ireland

- MSc in Computer Science – Course: *Mobile System Security*, Spring 2025 (Instructor: Paolo Palmieri). Prepared and delivered three lectures on Introduction to Cryptography and Firewalls.

Teaching Assistant & Guest Lecturer
Universidad Complutense de Madrid | Universidad Politécnica de Madrid, Madrid, Spain

- MSc in Formal Methods in Computer Science – Course: *Design and Analysis of Security Protocols*, Fall 2022–2023 (Instructors: Dario Fiore, Ignacio Cascudo, and Pedro Moreno-Sanchez). Prepared and delivered three lectures on Hash Functions, MACs, and Digital Signatures.

Teaching Assistant
EURECOM, Sophia Antipolis, France

- MSc/MEng/Post-Master programs – Course: *Secure Communications*, Fall 2015–2019 (Instructors: Refik Molva; Melek Önen). Instructed lab sessions on RSA encryption, RSA signatures, and Public Key Infrastructure (PKI).
- MSc/MEng/Post-Master programs – Course: *Security and Privacy for Big Data and Cloud*, Fall 2018 (Instructor: Melek Önen). Supervised student projects.
- MSc/MEng/Post-Master programs – Course: *Security Applications in Networking and Distributed Systems*, Spring 2015–2018 (Instructor: Refik Molva). Designed the Firewall lab and instructed sessions on Firewalls, and VPNs.

STUDENT ADVISING

PhD Co-advisorship

- Diego Castejon-Molina – PhD candidate, IMDEA Software Institute | Universidad Politécnica de Madrid, since Oct 2021 – Research: *Cryptographic Protocols for Secure and Privacy-preserving Payments in a Digital Economy* (Primary advisor: Pedro Moreno-Sanchez). Resulted in publication at PoPETs 2025 [2].

Master's Thesis Co-supervision

- Carlos Gonçalves da Silva – MSc, University of Luxembourg, since Feb 2026 – Research: *Hardware Integration of Zero-knowledge Succinct proofs* (Primary supervisor: Alex Biryukov).
- Javier Gomez-Martinez – MSc, Universidad Complutense de Madrid / Research Intern, IMDEA Software Institute, Feb–Jul 2023 – Research: *Algebraic Contingent Payments and Applications* (Primary supervisor: Dario Fiore). Resulted in publication at ACNS 2025 [1].

- Alberto del Amo Pastelero – MSc, Universidad Politécnica de Madrid / Research Intern, IMDEA Software Institute, Feb–Jul 2023 (thesis on hold due to personal reasons) – Research: *Cryptographic Protocols for Interoperability of Payment Channels* (Primary supervisor: Pedro Moreno-Sanchez).
- Ana-Marija Ereš – MSc, University of Zagreb / Research Intern, IMDEA Software Institute, Nov 2021–May 2022 – Research: *Analysis of the Impact of Collateral Funds in Payment Channel Networks* (Primary supervisor: Pedro Moreno-Sanchez). Resulted in publication in IEEE Access (Ereš & Bagić Babac, 2025).
- Duc Trinh – MEng Intern, EURECOM, Jul 2016–Feb 2017 – Topic: *Proofs of Retrievability with Secure Deduplication* (Primary supervisor: Melek Önen). Worked on H2020 TREDISEC project, implemented ML-PoR [7].

Semester Project Supervision

Typically ~2 months of full-time work spread across several months.

- Jeremy Brunet, Carlos Gonçalves da Silva, and Hasti Khajeh – MSc, University of Luxembourg, Fall 2025 – Topic: *Polynomial Commitment Schemes: KZG, ZK-KZG, and FRI*.
- Zahra Tahna and Maksym Voloshyn – MSc, University of Luxembourg, Fall 2025 – Topic: *MixBuy: Implementation of the Attestation Puzzle via Verifiable Witness Encryption*.
- Cedric Osornio-Gleanson – MSc, EURECOM, Spring 2018 – Topic: *Advanced Proofs of Data Reliability: Implementation of Linearly–homomorphic Tags*.
- Azim Hamadi and Ahmed Souissi – MEng, EURECOM, Fall 2017 – Topic: *Locally Recoverable Codes for Proofs of Data Reliability*.
- Arsham Aryaman and Meru Prabhat – MEng, EURECOM, Fall 2017 – Topic: *Firewall Laboratory*.
- Hamdi Ammar – MSc, EURECOM, Spring 2017 – Topic: *Proofs of Data Reliability: Experimental Evaluation of Time-constrained Proof Generation*.
- Linxia Gong and Yiyi Zhang – MEng, EURECOM, Fall 2016 – Topic: *Secure Deduplication: Implementation of PerfectDedup*.

TALKS

Invited Talks

- Lightning Network developers – Discreet Log Contracts (DLC) specification working group (Online) — Jun 2022
  Practical Decentralized Oracle Contracts
- Department of Electronic and Computing Engineering, Technical University of Crete, Chania, Greece — Sep 2017
  Privacy and Integrity Mechanisms for Cloud Storage

Conferences Talks

- 16th International Conference on Security and Cryptography, Prague, Czech Republic — Jul 2019
  PORTOS: Proof of Data Reliability for Real-World Distributed Outsourced Storage
- 6th International Workshop on Security on Cloud Computing, Incheon, Republic of Korea — Jun 2018
  POROS: Proof of Data Reliability for Outsourced Storage
- 2016 ACM on Cloud Security Workshop, Vienna, Austria — Oct 2016
  Message-Locked Proofs of Retrievability with Secure Deduplication

ACADEMIC SERVICE

Program Committee Membership

BCCA (2026); IEEE ICBC (2025 – 2026); CAAW @ FC (2025 – 2026); CBT @ ESORICS (2022 – 2026); CVCBT (2025); ACNS (2024); Tokenomics (2022 – 2023).

Journal Reviewing

Journal of Parallel and Distributed Computing; IEEE Transactions on Dependable and Secure Computing; Journal of Cloud Computing; International Journal of Information Security; IEEE Transactions on Information Forensics & Security.

Conference Reviewing (External Reviewer)

ESORICS (2017, 2025 – 2026); USENIX Security (2023); SECRYPT (2016 – 2017, 2023); CFS (2023); NDSS (2022); FC (2022); AFT (2022); IFIP SEC (2017, 2022); PST (2019); ACNS (2018); IEEE CNS (2018); ACM CCS (2017); CANS (2016 – 2017); ICDS (2016); ISC (2016); DPM (2015).

LANGUAGE SKILLS

English (Full professional proficiency) **French** (Fluent) **Greek** (Native speaker)

Last updated: June 3, 2026